



Microeconomics I Notes

- **Course:** Microeconomics I
- **Reference:** `Microeconomics_I_2026.pdf`
- **Source:** handwritten notes reorganized and cross-checked against the official lecture note

Diagrams and graph-only annotations are omitted. The remaining content is organized as a clean study note with definitions, propositions, examples, and derivations.

Contents

- 1. Consumer Theory and Welfare
- 2. Production Theory
- 3. Profit, Supply, and Cost
- 4. Exchange Economy and Competitive Equilibrium
- 5. Welfare Theorems and Production Economy

1. Consumer Theory and Welfare

Definition 1.1. Consumption Set

Example ($L = 2$).

$$X = \mathbb{R}_+^2$$
$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Definition 1.2. Preference Relation

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix} \in X$$

$x \succeq y \iff$ for consumer preference, x is at least as good as y .

$x \succ y \iff x$ is strictly better than y .

$x \sim y \iff x$ is indifferent to y .

Definition 1.3. Rational Preference

A rational preference satisfies:

$$\begin{cases} \text{complete,} \\ \text{transitive.} \end{cases}$$

Definition 1.4. Monotonicity

$\succeq$ is monotone if

$$y > x \implies y \succ x.$$

Here $y > x$ means $y \geq x$ and $y_j > x_j$ for some j . A weaker non-satiation form is $x \geq y \implies x \succeq y$.

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_L \end{bmatrix} \geq y = \begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix} \iff x_1 \geq y_1, \dots, x_L \geq y_L.$$

Definition 1.5. Convex Preference

The upper contour set is denoted by

$$UC(x)$$

Convex: $A \subset X$ is convex if

$$\forall a, a' \in A, \quad 0 \leq t \leq 1,$$

$$ta + (1 - t)a' \in A.$$

Additional continuity condition. The preference relation $\succeq$ is continuous if $UC(x)$ is closed and

$$LC(x) := \{y \in X \mid y \preceq x\}$$

is also closed.

Definition 1.6. Utility Function

A function $u : X \rightarrow \mathbb{R}$ represents the preference relation $\succeq$ if

$$x \succ y \iff u(x) > u(y), \quad x \sim y \iff u(x) = u(y).$$

Proposition 1.7. Utility Representation: Proof Notes

$$\forall x, y \in X \implies u(x), u(y) \in \mathbb{R}.$$

Then there are three possibilities:

$$\begin{cases} u_x > u_y, \\ u_y > u_x, \\ u_x = u_y. \end{cases}$$

Hence

$$\begin{cases} x \succ y, \\ y \succ x, \\ x \sim y, \end{cases} \implies \succeq \text{ is complete.}$$

If

$$x \succeq y, \quad y \succeq z,$$

then

$$u(x) \geq u(y) \geq u(z) \implies x \succeq z.$$

Proposition 1.8. Monotone Transformation: Proof Notes

$$\begin{aligned}x \succ y &\iff u(x) > u(y) \\&\iff \varphi(u(x)) > \varphi(u(y)) \\&\iff (\varphi \circ u)(x) > (\varphi \circ u)(y).\end{aligned}$$

Definition 1.9. Indifference Set

The indifference set through x is

$$I(x) = \{y \in X \mid y \sim x\}.$$

If the preference is represented by u , then

$$I(x) = \{y \in X \mid u(y) = u(x)\}.$$

Definition 1.10. Marginal Rate of Substitution

For a differentiable utility function u , the marginal rate of substitution between goods i and j is

$$\mu_{ij}(x) = \frac{\partial u(x)/\partial x_i}{\partial u(x)/\partial x_j}.$$

Example 1.12. Cobb-Douglas Utility

$$\varphi(u) = \log u.$$

$$(\varphi \circ u)(x) = \alpha \log x_1 + (1 - \alpha) \log x_2.$$

Example 1.13. CES Utility

$$u(x_1, x_2) = [x_1^\alpha + x_2^\alpha]^{1/\alpha}, \quad \alpha < 1, \quad \alpha \neq 0.$$

$$\varphi(u) = u^\alpha \implies (\varphi \circ u)(x) = x_1^\alpha + x_2^\alpha.$$

$$\mu = \frac{\alpha x_1^{\alpha-1}}{\alpha x_2^{\alpha-1}} = \left(\frac{x_2}{x_1}\right)^{1-\alpha}.$$

Proposition 1.15. CES Limit Cases

CES:

$$u = [x_1^\alpha + x_2^\alpha]^{1/\alpha}.$$

If

$$\alpha \rightarrow 1,$$

then

$$CES \rightarrow \text{linear}.$$

If

$$\alpha \rightarrow 0,$$

then MRS of CES:

$$\mu_{12} = \left(\frac{x_2}{x_1} \right)^{1-\alpha} \rightarrow \frac{x_2}{x_1},$$

form of Cobb-Douglas:

$$u = x_1^{1/2} x_2^{1/2}.$$

If

$$\alpha \rightarrow -\infty,$$

then

$$\begin{aligned} CES &= [x_1^\alpha + x_2^\alpha]^{1/\alpha}, \quad x_1 < x_2. \\ &= \left[x_1^\alpha \left(1 + \left(\frac{x_2}{x_1} \right)^\alpha \right) \right]^{1/\alpha} \\ &= x_1 \left[1 + \left(\frac{x_2}{x_1} \right)^\alpha \right]^{1/\alpha}. \end{aligned}$$

Since

$$\frac{x_2}{x_1} > 1, \quad \left(\frac{x_2}{x_1}\right)^\alpha \rightarrow 0 \quad \text{as } \alpha \rightarrow -\infty,$$

and

$$\frac{1}{\alpha} \rightarrow 0,$$

we get

$$\alpha \rightarrow -\infty, \quad x_1[1 + 0]^0 = x_1,$$

the smaller one in $[x_1, x_2]$.

Definitions 1.18-1.19. Consumer's Problem and Walrasian Demand

$$x(p, w) \in \arg \max u(x)$$

subject to

$$p \cdot x \leq w, \quad x \in X = \mathbb{R}_+^L.$$

Here $x(p, w)$ is a maximizer of u .

Example:

$$\arg \max(-x^2) \implies x = 0.$$

$$p \cdot x = p_1 x_1 + \cdots + p_L x_L.$$

If $L = 2$:

$$p_1 x_1 + p_2 x_2 \leq w.$$

Example:

$$u(x) = \alpha \log x_1 + (1 - \alpha) \log x_2.$$

$$\max u(x_1, x_2) \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 \leq w, \quad x_1, x_2 \geq 0.$$

$$\mathcal{L}(x_1, x_2, \lambda) = u(x) + \lambda(w - p_1 x_1 - p_2 x_2).$$

First-order conditions.

$$\frac{\partial \mathcal{L}}{\partial x_1} = 0, \quad \frac{\partial \mathcal{L}}{\partial x_2} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0.$$

Thus

$$\begin{aligned} \frac{\alpha}{x_1} - \lambda p_1 &= 0 \implies \alpha = \lambda p_1 x_1, \\ \frac{1 - \alpha}{x_2} - \lambda p_2 &= 0 \implies 1 - \alpha = \lambda p_2 x_2, \\ w - p_1 x_1 - p_2 x_2 &= 0. \end{aligned}$$

Therefore

$$\begin{aligned} \lambda &= \frac{1}{w}, \\ x_1 &= \frac{w\alpha}{p_1}, \quad x_2 = \frac{w(1 - \alpha)}{p_2}. \end{aligned}$$

Interpretation of λ .

Proposition 1.22. Properties of Walrasian Demand

Demand function properties:

$$\begin{cases} \text{homogeneity of degree zero: } x(\alpha p, \alpha w) = x(p, w), \\ \text{Walras' law: for all } p > 0, \quad p \cdot x(p, w) = w. \end{cases}$$

Definition 1.23. Indirect Utility Function

$$v(p, w) := \text{the max level of } u.$$

$$\implies u(x(p, w)) \quad \text{s.t. } p \cdot x \leq w.$$

In Cobb-Douglas,

$$v(p, w) = \alpha \log \frac{\alpha w}{p_1} + (1 - \alpha) \log \frac{(1 - \alpha)w}{p_2}.$$

D 1.25:

$$\lambda = \frac{\partial v(p, w)}{\partial w}.$$

Thus λ is the shadow price of wealth, or the marginal utility of wealth.

Cobb-Douglas demand formula.

$$u = x_1^\alpha x_2^\beta, \quad \alpha, \beta > 0,$$

or

$$u = \alpha \ln x_1 + \beta \ln x_2.$$

Then

$$\arg \max u$$

gives

$$x_1 = \frac{\alpha}{\alpha + \beta} \frac{w}{p_1}, \quad x_2 = \frac{\beta}{\alpha + \beta} \frac{w}{p_2}.$$

Here α, β seem to stand for the degree of preference for each good.

$$\frac{w}{p_i}$$

is the intercept.

Example: For CES max

$$u(x_1, x_2) = (x_1^\alpha + x_2^\alpha)^{1/\alpha}.$$

Let

$$\alpha = \frac{1}{2}.$$

Then

$$u = (\sqrt{x_1} + \sqrt{x_2})^2 \implies u = \sqrt{x_1} + \sqrt{x_2}.$$

$$\max u \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 \leq w.$$

$$\mathcal{L} = u + \lambda(w - p_1 x_1 - p_2 x_2).$$

FOCs.

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{1}{2\sqrt{x_1}} - \lambda p_1 = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = \frac{1}{2\sqrt{x_2}} - \lambda p_2 = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = w - p_1 x_1 - p_2 x_2 = 0.$$

Then

$$\frac{2\sqrt{x_2}}{2\sqrt{x_1}} = \frac{p_1}{p_2} \implies \sqrt{\frac{x_2}{x_1}} = \frac{p_1}{p_2} \implies x_2 = \left(\frac{p_1}{p_2}\right)^2 x_1.$$

Hence

$$x_1 = \frac{w}{p_1} \cdot \frac{p_2}{p_1 + p_2}, \quad x_2 = \frac{w}{p_2} \cdot \frac{p_1}{p_1 + p_2}.$$

Definition 1.25. Lagrange Multiplier and Shadow Price

λ 's meaning:

$$\lambda = \frac{\partial v(p, w)}{\partial w}.$$

Proof.

$$\begin{aligned} \frac{\partial v}{\partial w} &= \frac{\partial}{\partial w} [u(x_1(p, w), x_2(p, w))] \\ &= \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial w} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial w}. \end{aligned}$$

In the $\mathcal{L}$ function,

$$\frac{\partial u}{\partial x_1} = \lambda p_1, \quad \frac{\partial u}{\partial x_2} = \lambda p_2.$$

Therefore

$$\begin{aligned} &= \lambda p_1 \frac{\partial x_1}{\partial w} + \lambda p_2 \frac{\partial x_2}{\partial w} \\ &= \lambda \left\{ p_1 \frac{\partial x_1}{\partial w} + p_2 \frac{\partial x_2}{\partial w} \right\} \\ &= \lambda \frac{\partial}{\partial w} [p_1 x_1 + p_2 x_2] \\ &= \lambda \frac{\partial w}{\partial w} = \lambda. \end{aligned}$$

Example: Cobb-Douglas

$$p_1 \left(\frac{\alpha}{\alpha + \beta} \frac{w}{p_1} \right) + p_2 \left(\frac{\beta}{\alpha + \beta} \frac{w}{p_2} \right) = w.$$
$$\mathcal{L} = u(x) + \lambda \{w - p_1 x_1 - p_2 x_2\}.$$

Here λ is the shadow price of wealth, not utility from saving. The Lagrangian is used to solve the constrained utility maximization problem through the FOCs.

Example: Leontief

$$u = \min\{x_1, x_2\}.$$
$$\max u \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 \leq w.$$

Then

$$x_1 = x_2 \implies x_1 = x_2 = \frac{w}{p_1 + p_2}.$$

More generally,

$$u = \min\{\alpha x_1, \beta x_2\}.$$

Then

$$x_2 = \frac{\alpha}{\beta} x_1,$$

and

$$x_1 \left(p_1 + p_2 \frac{\alpha}{\beta} \right) = w.$$

Thus

$$x_1 = \frac{\beta}{\beta p_1 + \alpha p_2} w, \quad x_2 = \frac{\alpha}{\beta p_1 + \alpha p_2} w.$$

Here, indirect utility:

$$\begin{aligned} v(p, w) &= \min \left\{ \alpha \frac{\beta}{\beta p_1 + \alpha p_2} w, \beta \frac{\alpha}{\beta p_1 + \alpha p_2} w \right\} \\ &= \frac{\alpha \beta}{\beta p_1 + \alpha p_2} w. \end{aligned}$$

Hence

$$\lambda = \frac{\partial v}{\partial w} = \frac{\alpha \beta}{\beta p_1 + \alpha p_2}.$$

Definition 1.27. Expenditure Minimization Problem

Expenditure minimization:

$$\min p \cdot x$$

subject to

$$u(x) \geq u = \text{constant}.$$

Then

$$\mathcal{L} = p \cdot x + \lambda \{u - u(x_1, \dots, x_L)\}.$$

Example, $L = 2$:

$$u = x_1^\alpha x_2^{1-\alpha}.$$

$$\min p_1 x_1 + p_2 x_2$$

subject to

$$\alpha \log x_1 + (1 - \alpha) \log x_2 \geq \log u.$$

or

$$u(x_1, x_2) \geq u.$$

$$\mathcal{L} = p_1 x_1 + p_2 x_2 + \lambda \{\log u - \alpha \log x_1 - (1 - \alpha) \log x_2\}.$$

FOCs.

$$\frac{\partial \mathcal{L}}{\partial x_1} = p_1 - \frac{\lambda \alpha}{x_1} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = p_2 - \frac{\lambda(1 - \alpha)}{x_2} = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \log u - \alpha \log x_1 - (1 - \alpha) \log x_2 = 0.$$

Hence

$$\frac{p_1}{p_2} = \frac{\alpha x_2}{(1 - \alpha) x_1},$$

so

$$x_2 = \frac{p_1}{p_2} \cdot \frac{1 - \alpha}{\alpha} x_1.$$

Also

$$\begin{aligned} u &= x_1^\alpha x_2^{1-\alpha} = x_1^\alpha \left(\frac{1 - \alpha}{\alpha} \frac{p_1}{p_2} x_1 \right)^{1-\alpha} \\ &= \left(\frac{1 - \alpha}{\alpha} \frac{p_1}{p_2} \right)^{1-\alpha} x_1. \end{aligned}$$

Therefore

$$\begin{aligned} x_1 &= \left(\frac{\alpha}{1 - \alpha} \frac{p_2}{p_1} \right)^{1-\alpha} u = h_1(p_1, p_2, u), \\ x_2 &= \left(\frac{1 - \alpha}{\alpha} \frac{p_1}{p_2} \right)^\alpha u = h_2(p_1, p_2, u). \end{aligned}$$

Example 1.28. Hicksian Demand for Cobb-Douglas Utility

$$h(p, u) = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \implies \text{Hicksian demand function.}$$

Minimized value of expenditure:

$$\begin{aligned} & p_1 h_1(p_1, p_2, u) + p_2 h_2(p_1, p_2, u) \\ &= p_1 \left(\frac{\alpha}{1-\alpha} \frac{p_2}{p_1} \right)^{1-\alpha} u + p_2 \left(\frac{1-\alpha}{\alpha} \frac{p_1}{p_2} \right)^{\alpha} u \\ &= \left\{ \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} p_2^{1-\alpha} p_1^{\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} p_1^{\alpha} p_2^{1-\alpha} \right\} u \\ &= \left\{ \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} \right\} p_1^{\alpha} p_2^{1-\alpha} u \\ &= \left\{ \frac{\alpha}{1-\alpha} + 1 \right\} \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} p_1^{\alpha} p_2^{1-\alpha} u \\ &= (1-\alpha)^{\alpha-1} \alpha^{-\alpha} p_1^{\alpha} p_2^{1-\alpha} u \\ &= \left(\frac{p_1}{\alpha} \right)^{\alpha} \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u. \end{aligned}$$

Expenditure function:

$$e(p, u) = \left(\frac{p_1}{\alpha} \right)^{\alpha} \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u.$$

Proposition 1.29. Relation Between Expenditure Minimization and Utility Maximization

$$p_1 x_1 + p_2 x_2 = e(p, u_0).$$

$$u_0 = \text{constant.}$$

$$u(x_1^*, x_2^*) = v(p, w).$$

Corollary 1.30. Duality Between Expenditure and Indirect Utility

Duality:

$$e(p, u) \mid_{u=v(p, w)} = w.$$

$$v(p, w) \mid_{w=e(p, u)} = u.$$

$$x(p, w) \mid_{w=e(p, u)} = h(p, u).$$

$$h(p, u) \mid_{u=v(p, w)} = x(p, w).$$

Example:

$$u = x_1^\alpha x_2^{1-\alpha}.$$

1. Cobb-Douglas formula:

$$x_1 = \alpha \frac{w}{p_1}, \quad x_2 = (1 - \alpha) \frac{w}{p_2}.$$

2. By substitution:

$$v(p, w) = \left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{1 - \alpha}{p_2} \right)^{1-\alpha} w.$$

3. By the duality of the objective function:

$$v(p, e(p, u)) = u, \quad w = e.$$

Therefore

$$\left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{1 - \alpha}{p_2} \right)^{1-\alpha} e(p, u) = u,$$

so

$$e(p, u) = \left(\frac{p_1}{\alpha} \right)^\alpha \left(\frac{p_2}{1 - \alpha} \right)^{1-\alpha} u.$$

4. By the duality of solutions, get Hicksian demand function:

$$h_1(p, u) = x_1(p, e(p, u)) = \frac{\alpha e(p, u)}{p_1} = \left(\frac{\alpha}{1 - \alpha} \frac{p_2}{p_1} \right)^{1-\alpha} u.$$

$$h_2(p, u) = x_2(p, e(p, u)) = \frac{(1 - \alpha)e(p, u)}{p_2} = \left(\frac{1 - \alpha}{\alpha} \frac{p_1}{p_2} \right)^{\alpha} u.$$

Example:

$$u = x_1^{\alpha} x_2^{\beta} x_3^{\gamma}, \quad \alpha + \beta + \gamma = 1.$$

$$x_1 = \alpha \frac{w}{p_1}, \quad x_2 = \beta \frac{w}{p_2}, \quad x_3 = \gamma \frac{w}{p_3}.$$

$$v(p, w) = \left(\frac{\alpha}{p_1} \right)^{\alpha} \left(\frac{\beta}{p_2} \right)^{\beta} \left(\frac{\gamma}{p_3} \right)^{\gamma} w.$$

$$v(p, e(p, u)) = u.$$

Hence

$$e = \left(\frac{p_1}{\alpha} \right)^{\alpha} \left(\frac{p_2}{\beta} \right)^{\beta} \left(\frac{p_3}{\gamma} \right)^{\gamma} u.$$

Example:

$$\min rK + wL \quad \text{s.t.} \quad L^{\alpha} K^{1-\alpha} \geq y.$$

Equivalent to

$$\max L^{\alpha} K^{1-\alpha} \quad \text{s.t.} \quad wL + rK \leq C.$$

Then

$$L = \alpha \frac{C}{w}, \quad K = (1 - \alpha) \frac{C}{r}.$$

Substitution:

$$y = \left(\alpha \frac{C}{w} \right)^{\alpha} \left((1 - \alpha) \frac{C}{r} \right)^{1-\alpha} = \left(\frac{\alpha}{w} \right)^{\alpha} \left(\frac{1 - \alpha}{r} \right)^{1-\alpha} C.$$

By duality:

$$C = \left(\frac{w}{\alpha}\right)^\alpha \left(\frac{r}{1-\alpha}\right)^{1-\alpha} y.$$

Proposition 1.31. Properties of the Expenditure Function

The expenditure function $e(p, u)$:

$$\begin{cases} \text{HD 1 in } p : & e(\tau p, u) = \tau e(p, u), \\ \text{strictly increasing in } u \text{ and nondecreasing in } p, \\ \text{concave in } p, \\ \text{continuous in } (p, u). \end{cases}$$

$e(p, u)$ is said concave in p

if $p, p' \gg 0$,

$$e(tp + (1-t)p', u) \geq te(p, u) + (1-t)e(p', u).$$

Proof of concavity.

$$\begin{aligned} & e(tp + (1-t)p', u) \\ &= \{tp + (1-t)p'\} \cdot h(tp + (1-t)p', u) \\ &= tp \cdot h(tp + (1-t)p', u) + (1-t)p' \cdot h(tp + (1-t)p', u) \\ &\geq tp \cdot h(p, u) + (1-t)p' \cdot h(p', u) \\ &= te(p, u) + (1-t)e(p', u). \end{aligned}$$

Here $h(p, u)$ is the minimizer of price vector p , but $h(tp + (1-t)p')$ is not the minimizer.

Proposition 1.33. Shephard's Lemma

$$h_i(p, u) = \frac{\partial}{\partial p_i} e(p, u), \quad i = 1, 2.$$

Example:

$$e(p, u) = \left(\frac{p_1}{\alpha}\right)^\alpha \left(\frac{p_2}{1-\alpha}\right)^{1-\alpha} u.$$

$$\begin{aligned}
\frac{\partial e}{\partial p_1} &= \alpha \left(\frac{p_1}{\alpha} \right)^{\alpha-1} \frac{1}{\alpha} \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u \\
&= \left(\frac{\alpha}{p_1} \right)^{1-\alpha} \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u \\
&= \left(\frac{\alpha p_2}{p_1(1-\alpha)} \right)^{1-\alpha} u = h_1(p, u).
\end{aligned}$$

Proof.

$$e(p, u) = p_1 h_1(p, u) + p_2 h_2(p, u).$$

Fix

$$\bar{p} = p_0 \text{ constant.}$$

Let

$$\psi(p) := e(p, u) - \{p_1 h_1(p_0, u) + p_2 h_2(p_0, u)\}.$$

By definition,

$$\psi(p_0) = 0,$$

and for all p ,

$$\psi(p) \leq 0.$$

That is,

$$\psi(p) = \{p_1 h_1(p, u) + p_2 h_2(p, u)\} - \{p_1 h_1(p_0, u) + p_2 h_2(p_0, u)\}.$$

ψ is maximized at p_0 , hence

$$\psi'(p_0) = 0.$$

$$\frac{\partial \psi}{\partial p_i} = \frac{\partial e(p, u)}{\partial p_i} - h_i(p_0, u) = 0 \quad \text{at } p = p_0.$$

Therefore

$$\frac{\partial e(p_0, u)}{\partial p_i} = h_i(p_0, u), \quad \forall p_0.$$

which means

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u).$$

Corollary 1.34. Properties of Hicksian Demand

(i)

$$\frac{\partial h}{\partial p} = \text{Jacobian of } h = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = \begin{bmatrix} \frac{\partial h_1}{\partial p_1} & \frac{\partial h_1}{\partial p_2} \\ \frac{\partial h_2}{\partial p_1} & \frac{\partial h_2}{\partial p_2} \end{bmatrix}$$

is always negative semidefinite.

Def: A is a 2×2 matrix. A is NSD if for

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix},$$

we have

$$z^\top A z \leq 0.$$

Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Then

$$\begin{aligned} & [z_1, z_2] \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \\ &= [z_1, z_2] \begin{bmatrix} a_{11}z_1 + a_{12}z_2 \\ a_{21}z_1 + a_{22}z_2 \end{bmatrix} \\ &= z_1(a_{11}z_1 + a_{12}z_2) + z_2(a_{21}z_1 + a_{22}z_2) \\ &= a_{11}z_1^2 + (a_{12} + a_{21})z_1z_2 + a_{22}z_2^2. \end{aligned}$$

For example,

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \implies -z_1^2 - z_2^2 \leq 0.$$

For a symmetric 2×2 matrix,

$$A \text{ is NSD} \iff a_{11} \leq 0, \quad a_{22} \leq 0, \quad \det(A) \geq 0.$$

(ii)

$$\frac{\partial h}{\partial p} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Proof.

By Shephard's lemma,

$$h_i = \frac{\partial e}{\partial p_i}.$$

Hence

$$\frac{\partial h_i}{\partial p_j} = \frac{\partial}{\partial p_j} \left(\frac{\partial e}{\partial p_i} \right) = \frac{\partial^2 e}{\partial p_j \partial p_i}.$$

Thus

$$\frac{\partial h}{\partial p} = \begin{bmatrix} \frac{\partial^2 e}{\partial p_1^2} & \frac{\partial^2 e}{\partial p_1 \partial p_2} \\ \frac{\partial^2 e}{\partial p_2 \partial p_1} & \frac{\partial^2 e}{\partial p_2^2} \end{bmatrix}.$$

By P 1.31, e is concave and twice continuously differentiable, so it has an NSD Hessian.

Example:

$$f(x, y) = -x^2 - y^2.$$

$$D^2 f = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}, \quad -2 \leq 0, \quad 4 - 0 \geq 0.$$

Also,

$$h_i = \frac{\partial e}{\partial p_i} = \frac{\partial}{\partial p_i}(p_1 h_1 + p_2 h_2).$$

$$\implies p_1 \frac{\partial h_1}{\partial p_i} + p_2 \frac{\partial h_2}{\partial p_i} = 0.$$

Therefore

$$p_1 \frac{\partial h_i}{\partial p_1} + p_2 \frac{\partial h_i}{\partial p_2} = 0.$$

And same:

$$\begin{bmatrix} \frac{\partial h_1}{\partial p_1} & \frac{\partial h_1}{\partial p_2} \\ \frac{\partial h_2}{\partial p_1} & \frac{\partial h_2}{\partial p_2} \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Proposition 1.35. Slutsky Decomposition

Review.

$$x_i(p, w) = h_i(p, v(p, w))$$

$$h_i(p, u) = x_i(p, e(p, u))$$

$$\frac{\partial h_i}{\partial p_j}(p, u) = \frac{\partial x_i(p, e(p, u))}{\partial p_j} + \frac{\partial x_i(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_j}$$

$$= \frac{\partial x_i}{\partial p_j}(p, e(p, u)) + \frac{\partial x_i}{\partial w}(p, e(p, u)) h_j(p, u)$$

Since

$$e(p, v(p, w)) = w,$$

therefore

$$\left. \frac{\partial h_i}{\partial p_j}(p, u) \right|_{u=v(p, w)} = \frac{\partial x_i}{\partial p_j}(p, w) + \frac{\partial x_i}{\partial w}(p, w) x_j(p, w)$$

Slutsky's Decomposition:

$$= \frac{\partial x_i}{\partial p_j} + x_j \frac{\partial x_i}{\partial w}$$

or

$$\frac{\partial x_i}{\partial p_j} = \frac{\partial h_i}{\partial p_j} \Big|_{u=v(p,w)} - x_j \frac{\partial x_i}{\partial w}$$

where

$$\frac{\partial x_i}{\partial p_j}$$

is the price effect,

$$\frac{\partial h_i}{\partial p_j} \Big|_{u=v(p,w)}$$

is the substitution effect, keeping utility level unchanged, and

$$x_j \frac{\partial x_i}{\partial w}$$

is the income effect.

Matrix form:

$$\begin{bmatrix} \frac{\partial h_1}{\partial p_1} & \frac{\partial h_1}{\partial p_2} \\ \frac{\partial h_2}{\partial p_1} & \frac{\partial h_2}{\partial p_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial x_1}{\partial p_1} & \frac{\partial x_1}{\partial p_2} \\ \frac{\partial x_2}{\partial p_1} & \frac{\partial x_2}{\partial p_2} \end{bmatrix} + \begin{bmatrix} x_1 \frac{\partial x_1}{\partial w} & x_2 \frac{\partial x_1}{\partial w} \\ x_1 \frac{\partial x_2}{\partial w} & x_2 \frac{\partial x_2}{\partial w} \end{bmatrix}$$

$$\frac{\partial h}{\partial p} = \frac{\partial x}{\partial p} + \frac{\partial x}{\partial w} x^T$$

Definition.

$$S = \frac{\partial h}{\partial p} \Big|_{u=v(p,w)}$$

is the Slutsky substitution matrix.

Proposition 1.36. Properties of the Slutsky Matrix

1. $S^T = S$.
2. S is NSD.
3. $S p = 0$.

Proof.

1.

$$S = \frac{\partial h}{\partial p} = \frac{\partial^2 e}{\partial p \partial p} \Rightarrow \text{symmetric.}$$

2. See above.
3. See above.

Example 1:

$$\begin{cases} x_1 = aw - b_1 p_1 + c_1 p_2, \\ x_2 = aw - b_2 p_2 + c_2 p_1. \end{cases}$$

Then it is generally not rationalizable unless the Slutsky matrix satisfies symmetry, negative semidefiniteness, and $S p = 0$.

$$\begin{bmatrix} -b_1 & c_1 \\ c_2 & -b_2 \end{bmatrix} + \begin{bmatrix} ax_1 & ax_2 \\ ax_1 & ax_2 \end{bmatrix} = \begin{bmatrix} x_1 a - b_1 & x_2 a + c_1 \\ x_1 a + c_2 & x_2 a - b_2 \end{bmatrix}$$

The first matrix is $\frac{\partial x}{\partial p}$, the second is $\frac{\partial x}{\partial w} x^T$, and the sum is

$$\frac{\partial h}{\partial p} = S.$$

Does not satisfy above 3 conditions.

Therefore cannot find a utility function.

Example 2:

$$\begin{cases} x_1 = \frac{1}{2} \frac{w}{p_1}, \\ x_2 = \frac{1}{2} \frac{w}{p_2}. \end{cases}$$

$$S = \begin{bmatrix} -\frac{w}{2p_1^2} & 0 \\ 0 & -\frac{w}{2p_2^2} \end{bmatrix} + \begin{bmatrix} x_1 \frac{1}{2p_1} & x_2 \frac{1}{2p_1} \\ x_1 \frac{1}{2p_2} & x_2 \frac{1}{2p_2} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{w}{4p_1^2} & \frac{w}{4p_1 p_2} \\ \frac{w}{4p_1 p_2} & -\frac{w}{4p_2^2} \end{bmatrix}$$

Definition 1.39. Money Metric Indirect Utility

Definition 1.39

p, q : price vectors.

w : income.

$$\mu(p \mid q, w) := e(p, v(q, w))$$

is money metric indirect utility.

It means:

$$(q, w) \mapsto v(q, w),$$

and then

$$(p, w) \mapsto e(p, v(q, w)).$$

If

$$e(p, v(q, w)) > w,$$

then unhappy.

If

$$e(p, v(q, w)) < w,$$

then happy.

$$\mu(q) = e(q, v(q, w)) = w.$$

Remark 1.40

$$\mu(p) = e(p, v(q, w)).$$

$$\frac{\partial \mu(p)}{\partial p_i} = h_i(p, v(q, w)) = x_i(p, e(p, v(q, w))) = x_i(p, \mu(p)).$$

Example:

$$x_1 = \alpha \frac{w}{p_1}, \quad x_2 = (1 - \alpha) \frac{w}{p_2}.$$

Then

$$\begin{aligned} \frac{\partial \mu}{\partial p_1} &= x_1(p, \mu) = \alpha \frac{\mu}{p_1}, \\ \frac{\partial \mu}{\partial p_2} &= x_2(p, \mu) = (1 - \alpha) \frac{\mu}{p_2}. \end{aligned}$$

Boundary condition:

$$\mu(q) = w.$$

Assume that

$$\mu = f(p_1)g(p_2).$$

Then

$$\begin{aligned} f'(p_1)g(p_2) &= \alpha \frac{f(p_1)g(p_2)}{p_1}, \\ f(p_1)g'(p_2) &= (1 - \alpha) \frac{f(p_1)g(p_2)}{p_2}, \end{aligned}$$

and

$$f(q_1)g(q_2) = w.$$

Therefore

$$\frac{f'(p_1)}{f(p_1)} = \frac{\alpha}{p_1}, \quad \frac{g'(p_2)}{g(p_2)} = \frac{1 - \alpha}{p_2}.$$

Thus

$$\frac{d \ln f(p_1)}{dp_1} = \alpha \frac{d \ln p_1}{dp_1},$$
$$\frac{d \ln g(p_2)}{dp_2} = (1 - \alpha) \frac{d \ln p_2}{dp_2}.$$

Therefore

$$\ln f(p_1) = \ln p_1^\alpha + C_1 = \ln(C_1 p_1^\alpha),$$
$$\ln g(p_2) = \ln p_2^{1-\alpha} + C_2 = \ln(C_2 p_2^{1-\alpha}).$$

So

$$f(p_1) = C_1 p_1^\alpha, \quad g(p_2) = C_2 p_2^{1-\alpha},$$

and

$$\mu = C p_1^\alpha p_2^{1-\alpha}.$$

Since

$$\mu(q) = w,$$

we have

$$C = \frac{w}{q_1^\alpha q_2^{1-\alpha}}.$$

Therefore

$$\mu(p \mid q, w) = p_1^\alpha p_2^{1-\alpha} \cdot \frac{w}{q_1^\alpha q_2^{1-\alpha}}.$$

Also,

$$\mu(p) = e(p, u), \quad u = v(q, w).$$

Thus, for the standard Cobb-Douglas utility $u = x_1^\alpha x_2^{1-\alpha}$,

$$e(p, u) = \left(\frac{p_1}{\alpha} \right)^\alpha \left(\frac{p_2}{1-\alpha} \right)^{1-\alpha} u.$$

Since, under the standard normalization,

$$u = v(q, w) = \left(\frac{\alpha}{q_1} \right)^\alpha \left(\frac{1 - \alpha}{q_2} \right)^{1-\alpha} w,$$

then

$$u(p) = \frac{w}{p_1^\alpha p_2^{1-\alpha}}.$$

Since

$$x_1 = \alpha \frac{w}{p_1}, \quad x_2 = (1 - \alpha) \frac{w}{p_2},$$

we have

$$p_1 = \frac{\alpha w}{x_1}, \quad p_2 = \frac{w(1 - \alpha)}{x_2}.$$

Therefore

$$\begin{aligned} u &= \frac{w}{\left(\frac{\alpha w}{x_1} \right)^\alpha \left(\frac{(1 - \alpha)w}{x_2} \right)^{1-\alpha}} \\ &= \left(\frac{x_1}{\alpha} \right)^\alpha \left(\frac{x_2}{1 - \alpha} \right)^{1-\alpha} \sim x_1^\alpha x_2^{1-\alpha}. \end{aligned}$$

Definition.

$$\begin{aligned} p^0 &= (p_1^0, p_2^0), & t &= 0, \\ p^1 &= (p_1^1, p_2^1), & t &= 1. \end{aligned}$$

Assume

$$p_1^0 > p_1^1, \quad \bar{p}_2^0 = \bar{p}_2^1 = \bar{p}_2, \quad \bar{w} = \bar{w}.$$

Equivalent variation:

$$\begin{aligned} EV(p^0, p^1, w) &= \mu(p^0 \mid p^1, w) - w \\ &= e(p^0, v(p^1, w)) - w. \end{aligned}$$

Compensating variation:

$$CV(p^0, p^1, w) = w - e(p^1, v(p^0, w)).$$

If

$$CV = EV,$$

then

$$CV = EV = CS.$$

For equivalent variation,

$$\begin{aligned} EV(p^0, p^1, w) &= e(p_1^0, \bar{p}_2, v(p_1^1, \bar{p}_2, w)) - e(p_1^1, \bar{p}_2, v(p_1^1, \bar{p}_2, w)). \\ &= \int_{p_1^1}^{p_1^0} \frac{\partial e}{\partial p_1} (p_1, \bar{p}_2, v(p_1^1, \bar{p}_2, w)) dp_1 \\ &= \int_{p_1^1}^{p_1^0} h_1 (p_1, \bar{p}_2, v(p_1^1, \bar{p}_2, w)) dp_1. \end{aligned}$$

For compensating variation,

$$\begin{aligned} CV(p^0, p^1, w) &= e(p^0, v(p^0, w)) - e(p^1, v(p^0, w)). \\ &= \int_{p_1^1}^{p_1^0} \frac{\partial e}{\partial p_1} (p_1, \bar{p}_2, v(p^0, w)) dp_1 \\ &= \int_{p_1^1}^{p_1^0} h_1 (p_1, \bar{p}_2, v(p^0, \bar{p}_2, w)) dp_1. \end{aligned}$$

Thus

$$EV \geq CV.$$

Also,

$$0 > \frac{\partial h_i}{\partial p_i} = \frac{\partial x_i}{\partial p_i} + x_i \frac{\partial x_i}{\partial w} > \frac{\partial x_i}{\partial p_i}.$$

Slope of h is greater than slope of x .

Example:

$$u = \log x_1 + x_2, \quad p_2 = 1.$$

Maximize u , subject to

$$p_1 x_1 + x_2 \leq w.$$

Lagrangian:

$$\mathcal{L} = \log x_1 + x_2 + \lambda(w - p_1 x_1 - x_2).$$

FOCs.

$$\frac{1}{x_1} - \lambda p_1 = 0,$$

$$1 - \lambda = 0,$$

$$w - p_1 x_1 - x_2 = 0.$$

Therefore, for the interior solution with $w \geq 1$,

$$x_1 = \frac{1}{p_1}, \quad x_2 = w - 1.$$

$$v(p_1, w) = \log \left(\frac{1}{p_1} \right) + w - 1.$$

Thus

$$w = u + 1 + \log p_1 = e(p_1, u).$$

$$h_1 = \frac{\partial e}{\partial p_1} = \frac{1}{p_1} = x_1.$$

$$\begin{aligned} \mu(p \mid q, w) &= e(p, v(q, w)) \\ &= \left[\log \left(\frac{1}{q_1} \right) + w - 1 \right] + 1 + \log p_1 \\ &= w + \log \left(\frac{p_1}{q_1} \right). \end{aligned}$$

$$\begin{aligned}
EV &= e(p^0, v(p^1, w)) - w \\
&= \left[w + \log \left(\frac{p_1^0}{p_1^1} \right) \right] - w \\
&= \log \left(\frac{p_1^0}{p_1^1} \right). \\
CV &= w - e(p^1, v(p^0, w)) \\
&= w - \left[w + \log \left(\frac{p_1^1}{p_1^0} \right) \right] = \log \left(\frac{p_1^0}{p_1^1} \right).
\end{aligned}$$

If

$$EV = CV,$$

then

$$EV = CV = CS.$$

2. Production Theory

Production Set and Transformation Function

Definition 2.1. Production Vector

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_L \end{bmatrix} \in \mathbb{R}^L.$$

$y_i > 0$ output,

$y_i < 0$ input.

$$Y \subset \mathbb{R}^L$$

is the set of all feasible production $y \in \mathbb{R}^L$.

Remark 2.3. Standard Assumptions on the Production Set

1. $Y \neq \emptyset$.
2. Y is closed.
3. Inaction is possible: $0 \in Y$.
4. $Y \cap \mathbb{R}_+^L = \{0\}$. Therefore there is no $y > 0$ such that $y \in Y$: no free lunch.
5. $y \in Y, y' \leq y \Rightarrow y' \in Y$.
6. $y \in Y, y \neq 0 \Rightarrow -y \notin Y$.
7. $y, y' \in Y$ and $0 \leq \alpha \leq 1 \Rightarrow \alpha y + (1 - \alpha)y' \in Y$.

Transformation function notation:

$$\begin{aligned} F(y_1, y_2) &= 0, \\ y_2 &= f(y_1), \\ \bar{F}(y_1, f(y_1)) &= 0 \quad \forall y_1. \end{aligned}$$

Definition 2.4. Transformation Function

∂Y is the frontier, points of Y , the boundary set of Y .

$$y \in \partial Y \iff \bar{F}(y_1, \dots, y_L) = 0.$$

$$F : \mathbb{R}^L \rightarrow \mathbb{R}$$

is the transformation function.

If

$$y_1 \leq 0, \dots, y_{L-1} \leq 0$$

are inputs and

$$y_L \geq 0$$

is output, then there exists

$$y_L = f(y_1, \dots, y_{L-1})$$

such that

$$F(y_1, \dots, y_{L-1}, f(y_1, \dots, y_{L-1})) = 0$$

for all $y \in \partial Y$.

Example:

$$F(y_1, y_2) = 1 - y_1^2 - y_2^2 = 0.$$

$$y_2 = \sqrt{1 - y_1^2}.$$

$$\partial Y \iff F(y_1, \dots, y_{L-1}, y_L) = 0.$$

$$F(y_1, \dots, y_{L-1} \leq 0, y_L \geq 0) = 0 \Rightarrow y_L = f(y_1, \dots, y_{L-1}).$$

Definition 2.6. Production Function

$$z_1 = -y_1 \geq 0, \dots, z_{L-1} = -y_{L-1} \geq 0,$$

$$z_L = y_L \geq 0.$$

Therefore

$$z_L = f(z_1, \dots, z_{L-1}).$$

Example:

$$z_3 = z_1^\alpha z_2^\beta.$$

Equivalently,

$$y_3 = (-y_1)^\alpha (-y_2)^\beta.$$

Equivalently,

$$(-y_1)^\alpha (-y_2)^\beta - y_3 = 0.$$

Thus

$$\partial Y = \{(y_1, y_2, y_3) \in \mathbb{R}^3 \mid (-y_1)^\alpha (-y_2)^\beta - y_3 = 0\}.$$

Equivalently,

$$Y = \{(y_1, y_2, y_3) \in \mathbb{R}^3 \mid y_3 \leq (-y_1)^\alpha (-y_2)^\beta\}.$$

Definition 2.5. Marginal Rate of Transformation

$$MRT_{ij} = \frac{\frac{\partial \bar{F}}{\partial y_i}}{\frac{\partial \bar{F}}{\partial y_j}} = \frac{\frac{\partial f}{\partial z_i}}{\frac{\partial f}{\partial z_j}}, \quad z_L = f(z_1, \dots, z_{L-1}).$$

This is the marginal rate of transformation.

Example:

$$z_3 = z_1^\alpha z_2^\beta.$$

$$\frac{\frac{\partial f}{\partial z_1}}{\frac{\partial f}{\partial z_2}} = \frac{\alpha z_1^{\alpha-1} z_2^\beta}{\beta z_1^\alpha z_2^{\beta-1}} = \frac{\alpha}{\beta} \frac{z_2}{z_1} = \frac{\alpha}{\beta} \frac{y_2}{y_1}.$$

Also,

$$F(y_1, y_2, y_3) = (-y_1)^\alpha (-y_2)^\beta - y_3 = 0.$$

Then

$$\frac{\frac{\partial F}{\partial y_1}}{\frac{\partial F}{\partial y_2}} = \frac{-\alpha (-y_1)^{\alpha-1} (-y_2)^\beta}{-\beta (-y_1)^\alpha (-y_2)^{\beta-1}} = \frac{\alpha}{\beta} \frac{y_2}{y_1}.$$

In general,

$$F(y_1, \dots, y_{L-1}, f(y_1, \dots, y_{L-1})) = 0.$$

Therefore

$$\frac{\partial}{\partial y_i} [F(y_1, \dots, y_{L-1}, f(y_1, \dots, y_{L-1}))] = 0.$$

By chain rule,

$$\frac{\partial F}{\partial y_i} + \frac{\partial F}{\partial y_L} \cdot \frac{\partial f}{\partial y_i} = 0,$$

and

$$\frac{\partial F}{\partial y_j} + \frac{\partial F}{\partial y_L} \cdot \frac{\partial f}{\partial y_j} = 0.$$

Therefore

$$MRT = \frac{\frac{\partial F}{\partial y_i}}{\frac{\partial F}{\partial y_j}} = -\frac{\frac{\partial f}{\partial y_i}}{\frac{\partial f}{\partial y_j}} = \frac{\frac{\partial f}{\partial z_i}}{\frac{\partial f}{\partial z_j}}.$$

Profit Maximization:

$$z_L = f(z_1, \dots, z_{L-1}).$$

$$p = (p_1, \dots, p_L) \gg 0.$$

Profit:

$$\begin{aligned}\pi &= p_L z_L - p_1 z_1 - \dots - p_{L-1} z_{L-1} \\ &= p_L f(z_1, \dots, z_{L-1}) - p_1 z_1 - \dots - p_{L-1} z_{L-1}.\end{aligned}$$

Maximize over $z_1, \dots, z_{L-1}$:

$$\max_{z_1, \dots, z_{L-1}} p_L f(z_1, \dots, z_{L-1}) - (p_1 z_1 + \dots + p_{L-1} z_{L-1}).$$

FOCs.

$$p_L \frac{\partial f}{\partial z_i} - p_i = 0, \quad i \in [1, L-1].$$

Therefore

$$MRT = \frac{p_i}{p_j}.$$

Example:

$$z_3 = z_1^\alpha z_2^\beta, \quad \alpha + \beta < 1.$$

$$\pi = p_3 z_1^\alpha z_2^\beta - p_1 z_1 - p_2 z_2.$$

FOCs.

$$\frac{\partial \pi}{\partial z_1} = p_3 \alpha \frac{z_1^\alpha}{z_1} z_2^\beta - p_1 = 0.$$

$$\frac{\partial \pi}{\partial z_2} = p_3 \beta \frac{z_2^\beta}{z_2} z_1^\alpha - p_2 = 0.$$

Taking the ratio,

$$\frac{p_3 \alpha \frac{z_1^\alpha}{z_1} z_2^\beta}{p_3 \beta \frac{z_2^\beta}{z_2} z_1^\alpha} = \frac{p_1}{p_2}.$$

Thus

$$\frac{\alpha}{\beta} \frac{z_2}{z_1} = \frac{p_1}{p_2}.$$

Therefore

$$z_1 = \left[\alpha \left(\frac{\beta}{\alpha} \right)^\beta \frac{p_3}{p_1^{1-\beta} p_2^\beta} \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

$$z_2 = \left[\beta \left(\frac{\alpha}{\beta} \right)^\alpha \frac{p_3}{p_2^{1-\alpha} p_1^\alpha} \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

$$(z_1^*(p), z_2^*(p))$$

is the factor demand function.

$$z_3^*(p) = f(z_1^*(p), z_2^*(p))$$

is the supply function of good 3.

Then

$$z_3^*(p) = \left[\alpha \left(\frac{\beta}{\alpha} \right)^\beta \frac{p_3}{p_1^{1-\beta} p_2^\beta} \right]^{\frac{\alpha}{1-(\alpha+\beta)}} \left[\beta \left(\frac{\alpha}{\beta} \right)^\alpha \frac{p_3}{p_2^{1-\alpha} p_1^\alpha} \right]^{\frac{\beta}{1-(\alpha+\beta)}}.$$

Continuing.

$$z_3^*(p) = \left[\frac{\alpha^\alpha \beta^\beta p_3^{\alpha+\beta}}{p_1^\alpha p_2^\beta} \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

Equivalently,

$$z_3^*(p) = \left[\left(\frac{\alpha p_3}{p_1} \right)^\alpha \left(\frac{\beta p_3}{p_2} \right)^\beta \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

$\pi(p)$ = the maximized profit

$$= p_3 z_3^*(p) - p_1 z_1^*(p) - p_2 z_2^*(p)$$

$$= (1 - \alpha - \beta) \left[\left(\frac{\alpha}{p_1} \right)^\alpha \left(\frac{\beta}{p_2} \right)^\beta p_3 \right]^{\frac{1}{1-(\alpha+\beta)}}.$$

Review.

$$Y \subset \mathbb{R}^L$$

is the production set.

$$F : \mathbb{R}^L \rightarrow \mathbb{R}$$

is the transformation function.

$$y \in Y \iff \bar{F}(y) \leq 0.$$

$$y \in \partial Y \iff \bar{F}(y) = 0.$$

$$y_L \geq 0, \quad y_1 \leq 0, \dots, y_{L-1} \leq 0.$$

Therefore

$$z_L = y_L, \quad z_1 = -y_1 \geq 0, \dots, z_{L-1} = -y_{L-1} \geq 0.$$

$$F(y) = 0 \iff z_L = f(z_1, \dots, z_{L-1}).$$

3. Profit, Supply, and Cost

Definition.

$$\pi(p) = \max p \cdot y \quad \text{s.t. } y \in Y$$

profit function.

$$\begin{aligned} y(p) &= (y_1(p), \dots, y_L(p)) \\ &= \arg \max p \cdot y \quad \text{s.t. } y \in Y \end{aligned}$$

net supply function.

$$\therefore \pi(p) = p \cdot y(p) = \sum_{i=1}^L p_i y_i(p)$$

Proposition 2.9. Properties of the Profit Function

1. Homogeneous of degree one:

$$\pi(tp) = t\pi(p)$$

2. $\pi(p)$ is convex.
3. $y(p)$ is homogeneous of degree zero:

$$y(tp) = y(p)$$

4. Hotelling's lemma:

$$\frac{\partial}{\partial p_i} \pi(p) = y_i(p)$$

- 5.

$$\frac{\partial y_i}{\partial p_i} \geq 0, \quad \frac{\partial y_i}{\partial p_j} = \frac{\partial y_j}{\partial p_i}$$

Proof of Proposition 2.9.

- (ii) For all $p, p' \gg 0$, $0 \leq t \leq 1$,

$$\begin{aligned}
\pi(tp + (1-t)p') &= (tp + (1-t)p') \cdot y(tp + (1-t)p') \\
&= t[p \cdot y(tp + (1-t)p')] + (1-t)[p' \cdot y(tp + (1-t)p')] \\
&\leq t[p \cdot y(p)] + (1-t)[p' \cdot y(p')] \\
&= t\pi(p) + (1-t)\pi(p')
\end{aligned}$$

$$(iv) \ g : \mathbb{R}_{++}^L \rightarrow \mathbb{R},$$

$$g(p) = p \cdot y(p) - p \cdot y(p_0), \quad \forall p_0 \in \mathbb{R}_+^L$$

$$\Rightarrow g(p_0) = 0, \quad g'(p_0) \geq 0$$

$$\frac{\partial g(p_0)}{\partial p_i} = 0$$

$$\frac{\partial}{\partial p_i} [\bar{p} \cdot y(p) - p \cdot y(p_0)] = 0$$

$$\frac{\partial}{\partial p_i} \left[\sum_{j=1}^L \bar{p}_j y_j(p) - \sum_{j=1}^L p_j y_j(p_0) \right]$$

$$= y_i(p) + \sum_{j=1}^L p_j \frac{\partial y_j(p)}{\partial p_i} - y_i(p_0)$$

$$\Rightarrow \sum_{j=1}^L p_{0j} \frac{\partial y_j}{\partial p_i}(p_0) = 0 \quad \text{for all } p_0 \gg 0$$

$$\Rightarrow \sum_{j=1}^L p_j \frac{\partial y_j}{\partial p_i} \equiv 0$$

$$\therefore \frac{\partial}{\partial p_i} \pi(p) = \frac{\partial}{\partial p_i} \left(\sum_{j=1}^L p_j y_j(p) \right)$$

$$= y_i(p) + \sum_{j=1}^L p_j \frac{\partial y_j(p)}{\partial p_i} = y_i(p)$$

(v)

$$\frac{\partial y_i}{\partial p_i} = \frac{\partial}{\partial p_i} \left(\frac{\partial \pi}{\partial p_i} \right) \leftarrow \text{Hotelling's lemma}$$

$$\frac{\partial^2 \pi}{\partial p_i^2} \geq 0$$

because π is convex.

$$\frac{\partial y_i}{\partial p_j} = \frac{\partial y_j}{\partial p_i}$$

Example:

$$z_3 = z_1^\alpha + z_2^\alpha, \quad 0 < \alpha < 1$$

$$\pi = p_3(z_1^\alpha + z_2^\alpha) - p_1 z_1 - p_2 z_2$$

$$\frac{\partial \pi}{\partial z_1} = p_3 \alpha z_1^{\alpha-1} - p_1 = 0 \quad \Rightarrow \quad z_1 = \left(\frac{p_3 \alpha}{p_1} \right)^{\frac{1}{1-\alpha}}$$

$$\frac{\partial \pi}{\partial z_2} = p_3 \alpha z_2^{\alpha-1} - p_2 = 0 \quad \Rightarrow \quad z_2 = \left(\frac{p_3 \alpha}{p_2} \right)^{\frac{1}{1-\alpha}}$$

$$z_3 = \left(\frac{p_3 \alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{p_3 \alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}}$$

$$\pi(p) = p_3 \left[\left(\frac{p_3 \alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{p_3 \alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right] - p_1 \left(\frac{p_3 \alpha}{p_1} \right)^{\frac{1}{1-\alpha}} - p_2 \left(\frac{p_3 \alpha}{p_2} \right)^{\frac{1}{1-\alpha}}$$

$$= p_3^{\frac{1}{1-\alpha}} \left[\left(\frac{\alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{\alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right] - \frac{(\alpha p_3)^{\frac{1}{1-\alpha}}}{p_1^{\frac{\alpha}{1-\alpha}}} - \frac{(\alpha p_3)^{\frac{1}{1-\alpha}}}{p_2^{\frac{\alpha}{1-\alpha}}}$$

$$= p_3^{\frac{1}{1-\alpha}} \left[\left(\frac{\alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{\alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} - \alpha \left(\frac{\alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} - \alpha \left(\frac{\alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right]$$

$$= (1 - \alpha) p_3^{\frac{1}{1-\alpha}} \left[\left(\frac{\alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{\alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right]$$

And

$$y(p) = \left(- \left(\frac{\alpha p_3}{p_1} \right)^{\frac{1}{1-\alpha}}, - \left(\frac{\alpha p_3}{p_2} \right)^{\frac{1}{1-\alpha}}, \left(\frac{\alpha p_3}{p_1} \right)^{\frac{\alpha}{1-\alpha}}, + \left(\frac{\alpha p_3}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right)$$

$$\frac{\partial \pi}{\partial p_3} = (1 - \alpha) \times \frac{1}{1 - \alpha} p_3^{\frac{\alpha}{1-\alpha}} \left\{ \left(\frac{\alpha}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{\alpha}{p_2} \right)^{\frac{\alpha}{1-\alpha}} \right\}$$

$$= \left(\frac{\alpha p_3}{p_1} \right)^{\frac{\alpha}{1-\alpha}} + \left(\frac{\alpha p_3}{p_2} \right)^{\frac{\alpha}{1-\alpha}} = z_3$$

$$\frac{\partial y_1}{\partial p_2} = 0 = \frac{\partial y_2}{\partial p_1} \quad \text{in this case.}$$

$$\frac{\partial y_1}{\partial p_3} = \frac{\partial y_3}{\partial p_1}$$

Recovery of $\pi(p)$.

Example:

$$\pi = \frac{2p_3^2}{\sqrt{p_1 p_2}}$$

$$\Rightarrow z_3 = \frac{\partial \pi}{\partial p_3} = \frac{4p_3}{\sqrt{p_1 p_2}}$$

$$z_1 = -\frac{\partial \pi}{\partial p_1} = \frac{p_3^2}{p_1^{3/2} p_2^{1/2}}$$

$$z_2 = \frac{p_3^2}{p_1^{1/2} p_2^{3/2}}$$

$$z_1 z_2 = \frac{p_3^4}{p_1^2 p_2^2}$$

$$\Rightarrow z_3 = 4(z_1 z_2)^{1/4}$$

Cost minimization.

$$m = L - 1$$

$$z = (z_1, \dots, z_{L-1}, z_L) = (z_1, \dots, z_m, q)$$

$$p = (p_1, \dots, p_{L-1}, p_L) = (w_1, \dots, w_m, p)$$

Cost minimization problem, producer cost:

$$\min w \cdot z \quad \text{s.t. } f(z) \geq q$$

Profit maximization.

$$\max pq - c(w, q)$$

$$\Rightarrow q^* = q(w, p)$$

the supply function.

$$\Rightarrow \pi(w, p) = pq(w, p) - c(w, q(w, p))$$

profit function.

Example:

$$z_3 = z_1^\alpha z_2^\beta, \quad w_1 z_1 + w_2 z_2 = C$$

$$\Rightarrow z_1^* = \frac{\alpha}{\alpha + \beta} \cdot \frac{C}{w_1}$$

$$z_2^* = \frac{\beta}{\alpha + \beta} \cdot \frac{C}{w_2}$$

$$q = z_1^\alpha z_2^\beta = \frac{\alpha^\alpha \beta^\beta}{(\alpha + \beta)^{\alpha + \beta}} \cdot \frac{C^{\alpha + \beta}}{w_1^\alpha w_2^\beta}$$

$$\Rightarrow c(w_1, w_2, q)$$

Then substitute z_1, z_2 :

$$z_1(w, q), \quad z_2(w, q)$$

$$\pi = pq - c(w_1, w_2, q)$$

$$\frac{\partial \pi}{\partial q} \Rightarrow q^*(w_1, w_2, p)$$

Long run. If the production function has constant returns to scale, then:

$$q^*(w, p) = \begin{cases} +\infty, & p > \left(\frac{w_1}{\alpha}\right)^\alpha \left(\frac{w_2}{\beta}\right)^\beta, \\ [0, +\infty), & p = \left(\frac{w_1}{\alpha}\right)^\alpha \left(\frac{w_2}{\beta}\right)^\beta, \\ 0, & p < \left(\frac{w_1}{\alpha}\right)^\alpha \left(\frac{w_2}{\beta}\right)^\beta. \end{cases}$$

Proposition 2.12. Properties of the Cost Function

1.

$c(w, q)$ is homogeneous of degree 1 in w .

2.

$c(w, q)$ is concave in w .

3.

$z(w, q)$ is homogeneous of degree 0 in w .

4. Shephard's lemma:

$$\frac{\partial c}{\partial w_i} = z_i(w, q)$$

5.

$$\left[\frac{\partial z}{\partial w} \right] w = 0, \quad \text{i.e.} \quad \sum_{i=1}^m \frac{\partial z_j}{\partial w_i} w_i = 0 \quad \text{for each } j$$

6. If production function $f(z)$ is homogeneous of degree 1 (CRS),

$$\Rightarrow c(w, q) = c(w) \cdot q$$

unit cost (average cost).

7. If $f(z)$ is concave,

$$\Rightarrow c(w, q) \text{ is convex in } q.$$

For (6) proof. Assume $f(uz) = uf(z)$, $u > 0$.

$$c(w, q) = \min_z w \cdot z \quad \text{s.t.} \quad f(z) \geq q$$

$$\Rightarrow c(w, 1) = \min_z w \cdot z \quad \text{s.t.} \quad f(z) \geq 1$$

$$\begin{aligned}
c &= \min_{\bar{z}} q \cdot w\bar{z} \quad \text{s.t. } f(q\bar{z}) \geq q \\
&= q \left[\min_{\bar{z}} w\bar{z} \quad \text{s.t. } f(\bar{z}) \geq 1 \right] \\
&= q c(w, 1)
\end{aligned}$$

Also,

$$\begin{aligned}
c(w, q) &= \min_z w \cdot z \quad \text{s.t. } f(z) \geq q \\
&= \min_{\bar{z}} q w\bar{z} \quad \text{s.t. } f(q\bar{z}) \geq q \\
&= q \left[\min_{\bar{z}} w\bar{z} \quad \text{s.t. } f(\bar{z}) \geq 1 \right] \\
&= q c(w, 1)
\end{aligned}$$

Furthermore,

$$z_i(w, q) = \frac{\partial c(w)}{\partial w_i} \cdot q$$

a linear function of q .

$$\pi = (p - c(w))q = 0$$

Example:

$$\begin{aligned}
f(z_1, z_2) &= \min\{z_1, z_2\} \\
\min w \cdot z \quad \text{s.t. } f(z) &\geq q \\
\Rightarrow z_1 = z_2 &= q \\
c &= (w_1 + w_2)q
\end{aligned}$$

Problem 2.3. Find the supply function of

$$\begin{aligned}
q &= z_1^{1/2} z_2^{1/2} \\
\therefore \min w \cdot z \quad \text{s.t. } f(z) &= q \\
\Rightarrow \min w_1 z_1 + w_2 z_2 \quad \text{s.t. } z_1^{1/2} z_2^{1/2} &= q
\end{aligned}$$

$$\mathcal{L} = w_1 z_1 + w_2 z_2 + \lambda(q - z_1^{1/2} z_2^{1/2})$$

$$\frac{\partial \mathcal{L}}{\partial z_1} = w_1 - \frac{\lambda}{2} z_1^{-1/2} z_2^{1/2} = 0$$

$$\frac{\partial \mathcal{L}}{\partial z_2} = w_2 - \frac{\lambda}{2} z_2^{-1/2} z_1^{1/2} = 0$$

$$\frac{w_1}{w_2} = \frac{z_2}{z_1}$$

$$z_1 = z_2 \frac{w_2}{w_1}$$

$$q - \left(z_2 \frac{w_2}{w_1} \right)^{1/2} z_2^{1/2} = 0$$

$$q - z_2 \left(\frac{w_2}{w_1} \right)^{1/2} = 0$$

$$z_1 = \left(\frac{w_2}{w_1} \right)^{1/2} q$$

$$z_2 = \left(\frac{w_1}{w_2} \right)^{1/2} q$$

$$C = w_1 z_1 + w_2 z_2 = 2q(w_1 w_2)^{1/2}$$

$$\pi = pq - C = pq - 2q(w_1 w_2)^{1/2}$$

$$\frac{\partial \pi}{\partial q} = p - 2(w_1 w_2)^{1/2}$$

$$q = \begin{cases} +\infty, & p > 2(w_1 w_2)^{1/2}, \\ [0, \infty), & p = 2(w_1 w_2)^{1/2}, \\ 0, & p < 2(w_1 w_2)^{1/2}. \end{cases}$$

Problem 2.4. Find

$$q = [z_1^a + z_2^a]^{1/a}, \quad a < 1.$$

cost function and conditional factor demand.

Minimize

$$w_1 z_1 + w_2 z_2$$

$$q = [z_1^a + z_2^a]^{1/a}$$

$$q^a = [z_1^a + z_2^a]$$

$$\mathcal{L} = w_1 z_1 + w_2 z_2 + \lambda(q^a - z_1^a - z_2^a)$$

$$L_1 = w_1 - a\lambda z_1^{a-1} = 0$$

$$w_2 - a\lambda z_2^{a-1} = 0$$

$$\left(\frac{z_1}{z_2}\right)^{a-1} = \frac{w_1}{w_2}$$

$$\frac{z_1}{z_2} = \left(\frac{w_1}{w_2}\right)^{\frac{1}{a-1}}$$

$$z_1 = \left(\frac{w_1}{w_2}\right)^{\frac{1}{a-1}} z_2$$

$$q^a - \left(\frac{w_1}{w_2}\right)^{\frac{a}{a-1}} z_2^a - z_2^a = 0$$

$$z_2^a \left[1 + \left(\frac{w_1}{w_2}\right)^{\frac{a}{a-1}}\right] = q^a$$

$$z_2 = q \left[1 + \left(\frac{w_1}{w_2}\right)^{\frac{a}{a-1}}\right]^{-1/a}$$

$$z_1 = q \left[1 + \left(\frac{w_2}{w_1}\right)^{\frac{a}{a-1}}\right]^{-1/a}$$

$$c(w_1, w_2, q) = z_1 w_1 + z_2 w_2$$

$$= \left(\frac{w_1}{w_2}\right)^{\frac{1}{a-1}} z_2 w_1 + z_2 w_2$$

$$\begin{aligned}
&= z_2 w_2 \left[\left(\frac{w_1}{w_2} \right)^{\frac{a}{a-1}} + 1 \right] \\
&= q \left[1 + \left(\frac{w_1}{w_2} \right)^{\frac{a}{a-1}} \right]^{-\frac{1}{a}} \left[w_2 \left(1 + \left(\frac{w_1}{w_2} \right)^{\frac{a}{a-1}} \right) \right] \\
&= q \left[w_2^{\frac{a}{a-1}} + w_1^{\frac{a}{a-1}} \right]^{\frac{a-1}{a}}
\end{aligned}$$

Problem 2.6. Recovering the Production Function from Cost Derivatives

$$\begin{aligned}
\frac{\partial c}{\partial w_1} &= z_1 = \frac{1}{3} \left(\frac{w_2}{w_1} \right)^{2/3} q \\
\frac{\partial c}{\partial w_2} &= z_2 = \frac{2}{3} \left(\frac{w_1}{w_2} \right)^{1/3} q \\
\Rightarrow \frac{z_1}{z_2} &= \frac{1}{2} \frac{w_2}{w_1} \\
\Rightarrow z_1 &= \frac{q}{3} 2^{2/3} z_1^{2/3} z_2^{-2/3} \\
\Rightarrow q &= \frac{3}{2^{2/3}} z_1^{1/3} z_2^{2/3}
\end{aligned}$$

4. Exchange Economy and Competitive Equilibrium

Competitive Markets: Pure Exchange Economy

Pure exchange economy:

$$\mathcal{E} = \{(u^1, \omega^1), \dots, (u^I, \omega^I)\}$$

u^i is the utility function.

ω^i is the initial endowment.

$$\omega^i = \begin{pmatrix} \omega_1^i \\ \vdots \\ \omega_L^i \end{pmatrix} \in \mathbb{R}_+^L$$

$i \leftarrow \text{household}$

$j \leftarrow \text{goods}$

Step 1: Social planner's problem $\rightarrow U(\xi), \partial U(\xi)$.

Step 2: Pure exchange.

Step 3: Production economy.

Definition 3.2. Feasible Allocation

Allocation

$$A = (x^1, \dots, x^I)$$

is a vector of vectors,

$$x^i = \begin{pmatrix} x_1^i \\ \vdots \\ x_L^i \end{pmatrix}.$$

A is feasible in $\mathcal{E}$ if

$$\sum_{i=1}^I x^i \leq \sum_{i=1}^I \omega^i = \omega$$

aggregate initial endowment.

$$\bar{F} = \bar{F}(\mathcal{E})$$

is the set of feasible allocations A in $\mathcal{E}$.

$$= \left\{ (x^1, \dots, x^I) \mid \sum_i x^i \leq \omega \right\}$$

Definition 3.3. Pareto Optimal Allocation

For

$$A = (x^1, \dots, x^I) \in \bar{F}(\mathcal{E}),$$

$$u(A) = \begin{bmatrix} u^1(x^1) \\ \vdots \\ u^I(x^I) \end{bmatrix} \in \mathbb{R}^I.$$

$A \in \bar{F}(\mathcal{E})$ is Pareto optimal / efficient.

$$\iff \{ \bar{A} \in \bar{F}(\mathcal{E}) \mid u(\bar{A}) > u(A) \} = \emptyset$$

There's no other feasible allocation

$$\bar{A} = (\bar{x}^1, \dots, \bar{x}^I) \in \bar{F}(\mathcal{E})$$

s.t.

(a)

$$u^i(\bar{x}^i) \geq u^i(x^i) \quad \forall i = 1, \dots, I$$

(b)

$$u^j(\bar{x}^j) > u^j(x^j) \quad \exists j = 1, \dots, I.$$

Social planner's problem:

$$\mathcal{E} = \{(u^i, \omega^i)\} \Rightarrow P(u) = \text{the set of Pareto optimal allocations.}$$

Definition 3.4. Utility Possibility Set

Utility possibility set:

$$U = U(\mathcal{E})$$

$$= \{(u^1(x^1), \dots, u^I(x^I)) \mid A = (x^1, \dots, x^I) \in \bar{F}(\mathcal{E})\}$$

$$u(P(\mathcal{E})) \subset \partial U(\mathcal{E}) \quad \text{P 3.5.}$$

Example:

$$\mathcal{E} = \left\{ \left(\sqrt{x_1^1 x_2^1}, \omega^1 \right), \left(\sqrt{x_1^2 x_2^2}, \omega^2 \right) \right\}$$

Example:

$$A = \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) \in \bar{F}(\mathcal{E})$$

$$\omega^1 + \omega^2 = \omega = (1, 1)$$

$$\Rightarrow (u^1, u^2) = (1, 0) \in U(\mathcal{E})$$

$$A = \left(\begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \right) \in \bar{F}(\mathcal{E})$$

$$\Rightarrow (u^1, u^2) = \left(\frac{1}{2}, \frac{1}{2} \right) \in U(\mathcal{E})$$

If $A \in \bar{F}(\mathcal{E})$ is Pareto optimal,

$$\Rightarrow P(u) \subset \partial U(\mathcal{E})$$

Mathematically,

$$\max_{x^2} u^2(x^2) = \sqrt{x_1^2 x_2^2}$$

s.t.

$$\begin{cases} x_1^1 + x_1^2 \leq 1, \\ x_2^1 + x_2^2 \leq 1, \\ u^1(x^1) = \sqrt{x_1^1 x_2^1} \geq \xi. \end{cases}$$

$$\Rightarrow \max [u^2(x^2)]^2 = x_1^2 x_2^2 \quad \text{s.t.} \quad (1 - x_1^2)(1 - x_2^2) = \xi^2$$

$$\mathcal{L} = x_1^2 x_2^2 + \lambda \{(1 - x_1^2)(1 - x_2^2) - \xi^2\}$$

$$\frac{\partial \mathcal{L}}{\partial x_1^2} = x_2^2 - \lambda(1 - x_2^2) = 0$$

$$\Rightarrow \lambda = \frac{x_2^2}{1 - x_2^2} = \frac{x_1^2}{1 - x_1^2}$$

$$\frac{\partial \mathcal{L}}{\partial x_2^2} = x_1^2 - \lambda(1 - x_1^2) = 0$$

$$\Rightarrow x_2^2 = x_1^2 = C$$

$$(1 - C)^2 = \xi^2 \Rightarrow C = 1 - \xi \Rightarrow u^2 = 1 - \xi$$

Therefore, when $u^1 = \xi$, u^2 is at most $1 - \xi$.

$$U(\xi) = \{(u^1, u^2) \in \mathbb{R}_+^2 \mid u^1 + u^2 \leq 1\}$$

$$\partial[U(\xi)] = \{(u^1, u^2) \in \mathbb{R}_+^2 \mid u^1 + u^2 = 1\}$$

At a Pareto optimal allocation,

$$\frac{\partial u^i / \partial x_k^i}{\partial u^i / \partial x_l^i} = \frac{\partial u^h / \partial x_k^h}{\partial u^h / \partial x_l^h}, \quad i, h = 1, \dots, I, \quad k, l = 1, \dots, L.$$

Proof.

$$\max_{x^i} u^I(x^I)$$

subject to

$$\begin{cases} \sum_{i=1}^I x_1^i \leq \sum_{i=1}^I \omega_1^i, \\ \vdots \\ \sum_{i=1}^I x_L^i \leq \sum_{i=1}^I \omega_L^i, \\ u^1(x^1) \geq \xi^1, \dots, u^{I-1}(x^{I-1}) \geq \xi^{I-1}. \end{cases}$$

$$\mathcal{L} = u^I(x_1^I, \dots, x_L^I) + \lambda_1 \left(\sum_{i=1}^I \omega_1^i - \sum_{i=1}^I x_1^i \right) + \dots + \lambda_L \left(\sum_{i=1}^I \omega_L^i - \sum_{i=1}^I x_L^i \right) + \sum_{i=1}^{I-1} \mu_i (u^i(x^i) - \xi^i)$$

For I ,

$$\frac{\partial \mathcal{L}}{\partial x_l^I} = \frac{\partial u^I}{\partial x_l^I} - \lambda_l = 0.$$

Therefore,

$$\frac{\partial u^I / \partial x_l^I}{\partial u^I / \partial x_k^I} = \frac{\lambda_l}{\lambda_k}.$$

For $1 \leq i \leq I - 1$,

$$\frac{\partial \mathcal{L}}{\partial x_l^i} = 0 - \lambda_l + \mu_i \frac{\partial u^i}{\partial x_l^i} = 0.$$

Therefore,

$$\frac{\partial u^i / \partial x_l^i}{\partial u^i / \partial x_k^i} = \frac{\lambda_l}{\lambda_k}.$$

Hence,

$$\frac{\partial u^1 / \partial x_l^1}{\partial u^1 / \partial x_k^1} = \dots = \frac{\partial u^I / \partial x_l^I}{\partial u^I / \partial x_k^I}.$$

Example:

$$\mathcal{E} = \left\{ \left(\sqrt{x_1^1 x_2^1}, \omega^1 \right), \left(\sqrt{x_1^2 x_2^2}, \omega^2 \right) \right\},$$

$$\omega^1 + \omega^2 = (1, 1).$$

$$\frac{\partial u^1 / \partial x_1^1}{\partial u^1 / \partial x_2^1} = \frac{x_2^1}{x_1^1}, \quad \frac{\partial u^2 / \partial x_1^2}{\partial u^2 / \partial x_2^2} = \frac{x_2^2}{x_1^2}.$$

Therefore, for the Pareto optimal allocation

$$((x_1^1, x_2^1), (x_1^2, x_2^2))$$

satisfying

$$\begin{cases} \frac{x_2^1}{x_1^1} = \frac{x_2^2}{x_1^2}, \\ x_1^1 + x_1^2 = 1, \\ x_2^1 + x_2^2 = 1, \end{cases}$$

we have

$$\frac{x_2^1}{x_1^1} = \frac{1 - x_2^1}{1 - x_1^1}.$$

Therefore,

$$x_1^1 = x_2^1 = \xi.$$

Thus,

$$u^1 = \xi, \quad u^2 = 1 - \xi.$$

$$\partial[U(\xi)] = \{(\xi, 1 - \xi) : 0 \leq \xi \leq 1\}.$$

The Pareto optimal allocation:

$$P(\xi) = \left\{ \left(\begin{pmatrix} \xi \\ \xi \end{pmatrix}, \begin{pmatrix} 1 - \xi \\ 1 - \xi \end{pmatrix} \right) \mid 0 \leq \xi \leq 1 \right\}.$$

Example 2:

$$\mathcal{E} = \left\{ \left((\sqrt{x_1^1} + \sqrt{x_2^1})^2, \omega^1 \right), \left((\sqrt{x_1^2} + \sqrt{x_2^2})^2, \omega^2 \right) \right\}.$$

$$\omega^1 + \omega^2 = (1, 1).$$

$$\frac{\partial u^1 / \partial x_1^1}{\partial u^1 / \partial x_2^1} = \sqrt{\frac{x_2^1}{x_1^1}}, \quad \frac{\partial u^2 / \partial x_1^2}{\partial u^2 / \partial x_2^2} = \sqrt{\frac{x_2^2}{x_1^2}}.$$

Therefore,

$$\begin{cases} \sqrt{\frac{x_2^1}{x_1^1}} = \sqrt{\frac{x_2^2}{x_1^2}}, \\ x_1^1 + x_1^2 = 1, \\ x_2^1 + x_2^2 = 1. \end{cases}$$

Hence,

$$\frac{x_2^1}{x_1^1} = \frac{1 - x_2^1}{1 - x_1^1} \Rightarrow x_1^1 = x_2^1 = \xi \in [0, 1].$$

Therefore,

$$u^1 = 4\xi \in [0, 4].$$

$$u^2 = 4(1 - \xi) \in [0, 4].$$

$$P(\xi) = \left\{ \left(\begin{pmatrix} \xi \\ \xi \end{pmatrix}, \begin{pmatrix} 1 - \xi \\ 1 - \xi \end{pmatrix} \right) \mid 0 \leq \xi \leq 1 \right\}.$$

$$\partial[U(\xi)] = \{(u^1, u^2) \in \mathbb{R}_+^2 \mid u^1 + u^2 = 4\}.$$

Definition 3.9. Competitive Equilibrium Allocation

$A^* = (x_*^1, \dots, x_*^I) \in \bar{F}(\mathcal{E})$ together with p^* constitutes a competitive equilibrium if

(i)

$$x_*^i \in \arg \max u^i(x), \quad \text{s.t.} \quad p^* \cdot x \leq p^* \cdot \omega^i,$$

with

$$p^* \in \mathbb{R}_+^L.$$

(ii)

$$\sum_{i=1}^I x_*^i = \sum_{i=1}^I \omega^i.$$

Example:

$$\mathcal{E} = \left\{ \left(\sqrt{x_1^1 x_2^1}, (2, 1) \right), \left(\sqrt{x_1^2 x_2^2}, (1, 2) \right) \right\}.$$

$$\frac{\frac{1}{2x_1^1} \sqrt{x_1^1 x_2^1}}{\frac{1}{2x_2^1} \sqrt{x_1^1 x_2^1}} = \frac{x_2^1}{x_1^1}.$$

Therefore,

$$\frac{x_2^1}{x_1^1} = \frac{x_2^2}{x_1^2}, \quad x_1^1 + x_1^2 = 3, \quad x_2^1 + x_2^2 = 3.$$

Thus,

$$\frac{x_2^1}{x_1^1} = \frac{3 - x_2^1}{3 - x_1^1} \iff x_1^1 = x_2^1 = \xi.$$

Therefore,

$$P(\xi) = \left\{ \left(\begin{pmatrix} \xi \\ \xi \end{pmatrix}, \begin{pmatrix} 3 - \xi \\ 3 - \xi \end{pmatrix} \right) \mid 0 \leq \xi \leq 3 \right\}.$$

Competitive demands. Given $p = (p_1, p_2)$,

$$\max \sqrt{x_1^1 x_2^1} \quad \text{s.t.} \quad p_1 x_1^1 + p_2 x_2^1 \leq p_1 \cdot 2 + p_2 \cdot 1.$$

$$x_1^*(p) = \left(\frac{1}{2} \cdot \frac{2p_1 + p_2}{p_1}, \frac{1}{2} \cdot \frac{2p_1 + p_2}{p_2} \right) = \left(1 + \frac{1}{2} \frac{p_2}{p_1}, \frac{p_1}{p_2} + \frac{1}{2} \right).$$

$$\max \sqrt{x_1^2 x_2^2} \quad \text{s.t.} \quad p_1 x_1^2 + p_2 x_2^2 \leq p_1 \cdot 1 + p_2 \cdot 2.$$

$$x_2^*(p) = \left(\frac{1}{2} + \frac{p_2}{p_1}, \frac{p_1}{2p_2} + 1 \right).$$

$$x_1^* + x_2^* = \left(\frac{3}{2} + \frac{3}{2} \frac{p_2}{p_1}, \frac{3}{2} \frac{p_1}{p_2} + \frac{3}{2} \right).$$

This should equal

$$\omega^1 + \omega^2 = (2, 1) + (1, 2) = (3, 3).$$

Therefore,

$$\begin{cases} \frac{3}{2} + \frac{3}{2} \frac{p_2}{p_1} = 3, \\ \frac{3}{2} \frac{p_1}{p_2} + \frac{3}{2} = 3, \end{cases}$$

hence

$$p_1 = p_2.$$

Therefore, the equilibrium price ratio is determined here:

$$\frac{p_1^*}{p_2^*} = 1.$$

Competitive allocation:

$$x_*^1 = \left(\frac{3}{2}, \frac{3}{2} \right), \quad x_*^2 = \left(\frac{3}{2}, \frac{3}{2} \right).$$
$$u_*^1 = \frac{3}{2}, \quad u_*^2 = \frac{3}{2}.$$

Definition 3.10. Excess Demand Function

$$\zeta : \mathbb{R}_+^L \rightarrow \mathbb{R}^L, \quad p \mapsto \zeta(p),$$

$$\zeta(p) = \sum_{i=1}^I x^{i*}(p) - \sum_{i=1}^I \omega^i.$$

The first term is aggregate demand, and the second term is aggregate endowment.

Then p^* is equilibrium price iff

$$\zeta(p^*) = 0.$$

Theorem 3.11. Properties of Excess Demand

(i) $\zeta(p)$ is homogeneous of degree zero: for $\forall t > 0$,

$$\zeta(tp) = \zeta(p).$$

(ii) Walras' Law:

$$p \cdot \zeta(p) = \sum_{l=1}^L p_l \zeta_l(p) \equiv 0, \quad \forall p \in \mathbb{R}_+^L.$$

Proof of (i):

$$x^{i*}(p)$$

is homogeneous of degree zero because

$$x^{i*}(tp) \in \arg \max u^i(x) \quad \text{s.t.} \quad tp \cdot x \leq tp \cdot \omega^i.$$

Since $t > 0$,

$$x^{i*}(tp) \in \arg \max u^i(x) \quad \text{s.t.} \quad p \cdot x \leq p \cdot \omega^i.$$

Therefore,

$$\zeta(tp) = \sum_{i=1}^I x^{i*}(tp) - \sum_{i=1}^I \omega^i = \sum_{i=1}^I x^{i*}(p) - \sum_{i=1}^I \omega^i = \zeta(p).$$

Proof of (ii):

Since $u^i(x)$ is strictly increasing,

$$p \cdot x^{i*}(p) \equiv p \cdot \omega^i, \quad \forall p \in \mathbb{R}_+^L.$$

Therefore,

$$\sum_{i=1}^I p \cdot x^{i*}(p) = \sum_{i=1}^I p \cdot \omega^i.$$

Thus,

$$p \cdot \left[\sum_{i=1}^I (x^{i*}(p) - \omega^i) \right] = p \cdot \zeta(p) = 0.$$

In the case of $L = 2$,

$$p \cdot \zeta(p) = p_1 \zeta_1(p) + p_2 \zeta_2(p) \equiv 0, \quad \forall p \gg 0.$$

$$\Longleftrightarrow \zeta_1(p) = 0 \quad \text{iff} \quad \zeta_2(p) = 0.$$

And if

$$\zeta_1(p) > 0 \Longleftrightarrow \zeta_2(p) < 0,$$

$$\zeta_1(p) < 0 \Longleftrightarrow \zeta_2(p) > 0.$$

$$\zeta(p^*) = 0 \Longleftrightarrow \begin{cases} \zeta_1(p^*) = 0, \\ \zeta_2(p^*) = 0 \end{cases} \Rightarrow \zeta_1(p^*) = 0.$$

$$\Longleftrightarrow \zeta_1(1 \cdot p_1^*, 1 \cdot p_2^*) = 0.$$

$$\Rightarrow \zeta_1\left(\frac{p_1^*}{p_2^*}, 1\right) = 0, \quad t = \frac{1}{p_2^*} > 0.$$

Therefore,

$$\frac{p_1^*}{p_2^*}$$

matters, and is determined by the market as the equilibrium.

Example:

$$\mathcal{E} = \{(u^i, \omega^i), i = 1, \dots, I\}.$$

$$L = 3.$$

$$u^i(x_1, x_2, x_3) = x_1^\alpha x_2^\beta x_3^\gamma, \quad \alpha + \beta + \gamma = 1.$$

$$\omega^i = (\omega_1^i, \omega_2^i, \omega_3^i) \in \mathbb{R}_+^3.$$

For agent i ,

$$x^{i*}(p) = \left(\alpha \frac{p \cdot \omega^i}{p_1}, \beta \frac{p \cdot \omega^i}{p_2}, \gamma \frac{p \cdot \omega^i}{p_3} \right).$$

$$\zeta(p) = \sum_{i=1}^I \left(\alpha \frac{p \cdot \omega^i}{p_1}, \beta \frac{p \cdot \omega^i}{p_2}, \gamma \frac{p \cdot \omega^i}{p_3} \right) - \sum_{i=1}^I \omega^i.$$

Let

$$\omega = \sum_{i=1}^I \omega^i = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}.$$

Then

$$\zeta(p) = \left(\alpha \frac{p \cdot \omega}{p_1}, \beta \frac{p \cdot \omega}{p_2}, \gamma \frac{p \cdot \omega}{p_3} \right) - \omega.$$

$$\zeta(p) = (0, 0, 0)$$

if and only if

$$\begin{cases} \alpha \frac{p \cdot \omega}{p_1} = \omega_1, \\ \beta \frac{p \cdot \omega}{p_2} = \omega_2, \\ \gamma \frac{p \cdot \omega}{p_3} = \omega_3. \end{cases}$$

Equivalently,

$$\begin{cases} \alpha p \cdot \omega = p_1 \omega_1, \\ \beta p \cdot \omega = p_2 \omega_2, \\ \gamma p \cdot \omega = p_3 \omega_3. \end{cases}$$

Therefore,

$$p_1 \omega_1 : p_2 \omega_2 : p_3 \omega_3 = \alpha : \beta : \gamma.$$

$$p_1^* : p_2^* : p_3^* = \frac{\alpha}{\omega_1} : \frac{\beta}{\omega_2} : \frac{\gamma}{\omega_3}.$$

Endowments and preferences determine relative prices, not the absolute price level.

Example:

$$L = 2, \quad u^1 = u^2 = \min\{x_1, x_2\}.$$

$$\omega^1 = (\omega_1^1, \omega_2^1), \quad \omega^2 = (\omega_1^2, \omega_2^2).$$

$$\omega = \omega^1 + \omega^2 = (\omega_1, \omega_2).$$

Agent 1 income:

$$p \cdot \omega^1 = p_1 \omega_1^1 + p_2 \omega_2^1.$$

$$x^1(p) = \left(\frac{p \cdot \omega^1}{p_1 + p_2}, \frac{p \cdot \omega^1}{p_1 + p_2} \right).$$

Agent 2:

$$x^2(p) = \left(\frac{p \cdot \omega^2}{p_1 + p_2}, \frac{p \cdot \omega^2}{p_1 + p_2} \right).$$

$$\begin{aligned} \zeta(p) &= x^1(p) + x^2(p) - \omega \\ &= \left(\frac{p \cdot (\omega^1 + \omega^2)}{p_1 + p_2}, \frac{p \cdot (\omega^1 + \omega^2)}{p_1 + p_2} \right) - (\omega_1, \omega_2) \\ &= \left(\frac{p \cdot \omega}{p_1 + p_2} - \omega_1, \frac{p \cdot \omega}{p_1 + p_2} - \omega_2 \right) = 0. \end{aligned}$$

Therefore,

$$\begin{cases} \frac{p \cdot \omega}{p_1 + p_2} = \omega_1, \\ \frac{p \cdot \omega}{p_1 + p_2} = \omega_2. \end{cases}$$

If $\omega_1 \neq \omega_2$, no equilibrium.

If $\omega_1 = \omega_2 = \alpha$,

$$\frac{p_1 \alpha + p_2 \alpha}{p_1 + p_2} = \alpha.$$

Then any strictly positive price vector clears the market.

Theorem 3.13. Existence of Competitive Equilibrium

Competitive equilibrium of $\mathcal{E}$ exists if the consumption space X is bounded and closed, and

(i) u^i is strictly increasing.

(ii) u^i is strictly quasi-concave.

(iii) $\zeta(p)$ is continuous on $\mathbb{R}_+^L$.

For Leontief, (i) and (ii) are violated.

Roy's identity.

$V(m, p_1, p_2)$: indirect utility.

$$x_1(m, p_1, p_2) = - \frac{\frac{\partial V}{\partial p_1}}{\frac{\partial V}{\partial m}}.$$

Proof.

$$\max u(x) \quad \text{s.t.} \quad p \cdot x \leq m.$$

$$\mathcal{L} = u(x) + \lambda(m - p \cdot x).$$

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u}{\partial x_i} - \lambda p_i = 0.$$

$$V(m, p_1, p_2) = u(x_1(m, p_1, p_2), x_2(m, p_1, p_2)).$$

$$\begin{aligned} \frac{\partial V}{\partial p_1} &= \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial p_1} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial p_1} \\ &= \lambda p_1 \frac{\partial x_1}{\partial p_1} + \lambda p_2 \frac{\partial x_2}{\partial p_1} \\ &= \lambda \left\{ p_1 \frac{\partial x_1}{\partial p_1} + p_2 \frac{\partial x_2}{\partial p_1} \right\} \\ &= \frac{\partial V}{\partial m} \left\{ \frac{\partial}{\partial p_1} [p_1 x_1 + p_2 x_2] - x_1 \right\}. \end{aligned}$$

Since

$$p_1 x_1 + p_2 x_2 = m,$$

we have

$$\frac{\partial V}{\partial p_1} = - \frac{\partial V}{\partial m} x_1.$$

Therefore,

$$x_1 = - \frac{\frac{\partial V}{\partial p_1}}{\frac{\partial V}{\partial m}}.$$

5. Welfare Theorems and Production Economy

Review. Walrasian Equilibrium

$$\mathcal{E} = \{(u^i, \omega^i) : i = 1, \dots, I\}.$$

$A^* = (x_*^1, \dots, x_*^I) \in \bar{F}(\mathcal{E})$ and $p^* \in \mathbb{R}_{++}^L$ constitute a Walrasian equilibrium if

(i)

$$x_*^i \in \arg \max u^i(x) \quad \text{s.t.} \quad p^* \cdot x \leq p^* \cdot \omega^i, \quad \forall i;$$

(ii)

$$\sum_{i=1}^I x_*^i = \sum_{i=1}^I \omega^i = \omega.$$

Proposition 3.14. First Welfare Theorem

If u^i is strictly increasing and $A^* = (x_*^1, \dots, x_*^I)$ is a Walrasian equilibrium allocation with price p^* , then A^* is Pareto optimal:

$$\{A \in \bar{F}(\mathcal{E}) \mid u(A) > u(A^*)\} = \emptyset.$$

Proposition 3.15. Second Welfare Theorem

If u^i is strictly increasing and strictly quasi-concave, then for every $A^* \in P(\mathcal{E})$, there exist $\bar{\omega} = (\bar{\omega}^1, \dots, \bar{\omega}^I)$ and $p^* \in \mathbb{R}_{++}^L$ such that A^* is a Walrasian equilibrium of

$$\bar{\mathcal{E}} = \{(u^i, \bar{\omega}^i) : i = 1, \dots, I\}.$$

Example. Cobb-Douglas Exchange Economy

$$\mathcal{E} = \{(u^i, \omega^i) : i = 1, 2\}, \quad u^i(x^i) = (x_1^i)^\alpha (x_2^i)^{1-\alpha}, \quad \omega = \omega^1 + \omega^2.$$

Find A^* and show $(u^1(x_*^1), u^2(x_*^2)) \in \partial U(\mathcal{E})$.

$$x_1^i = \alpha \frac{p \cdot \omega^i}{p_1}, \quad x_2^i = (1 - \alpha) \frac{p \cdot \omega^i}{p_2}.$$

Market clearing:

$$\alpha \frac{p \cdot \omega}{p_1} = \omega_1 \implies \frac{p_2}{p_1} = \frac{1 - \alpha}{\alpha} \cdot \frac{\omega_1}{\omega_2}.$$

$$x_1^1 = \alpha \omega_1^1 + (1 - \alpha) \frac{\omega_1}{\omega_2} \omega_2^1, \quad x_2^1 = \alpha \frac{\omega_2}{\omega_1} \omega_1^1 + (1 - \alpha) \omega_2^1.$$

$$x_1^2 = \alpha \omega_1^2 + (1 - \alpha) \frac{\omega_1}{\omega_2} \omega_2^2, \quad x_2^2 = \alpha \frac{\omega_2}{\omega_1} \omega_1^2 + (1 - \alpha) \omega_2^2.$$

$$u^1(x_*^1) = \omega_1^\alpha \omega_2^{1-\alpha} \left(\alpha \frac{\omega_1^1}{\omega_1} + (1 - \alpha) \frac{\omega_2^1}{\omega_2} \right),$$

$$u^2(x_*^2) = \omega_1^\alpha \omega_2^{1-\alpha} \left(\alpha \frac{\omega_1^2}{\omega_1} + (1 - \alpha) \frac{\omega_2^2}{\omega_2} \right).$$

Pareto optimal allocations satisfy

$$\frac{x_2^1}{x_1^1} = \frac{x_2^2}{x_1^2} \implies \frac{x_1^1}{x_2^1} = \frac{\omega_1}{\omega_2}.$$

Contract curve:

$$x^1 = (\tau \omega_1, \tau \omega_2), \quad x^2 = ((1 - \tau) \omega_1, (1 - \tau) \omega_2), \quad \tau \in [0, 1].$$

$$u^1 + u^2 = \omega_1^\alpha \omega_2^{1-\alpha},$$

$$U(\mathcal{E}) = \{(u^1, u^2) \in \mathbb{R}_+^2 \mid u^1 + u^2 \leq \omega_1^\alpha \omega_2^{1-\alpha}\},$$

$$\partial U(\mathcal{E}) = \{(u^1, u^2) \in \mathbb{R}_+^2 \mid u^1 + u^2 = \omega_1^\alpha \omega_2^{1-\alpha}\}.$$

At A^* , since $\omega_1^1 + \omega_1^2 = \omega_1$ and $\omega_2^1 + \omega_2^2 = \omega_2$,

$$u^1(x_*^1) + u^2(x_*^2) = \omega_1^\alpha \omega_2^{1-\alpha}.$$

For $\beta \in [0, 1]$, let $\bar{\omega}^1 = \beta\omega$, $\bar{\omega}^2 = (1 - \beta)\omega$. Then

$$u^1 = \beta\omega_1^\alpha\omega_2^{1-\alpha}, \quad u^2 = (1 - \beta)\omega_1^\alpha\omega_2^{1-\alpha}.$$

Proof of Proposition 3.14. First Welfare Theorem

Let $A^* = (x_*^1, \dots, x_*^I)$ be a Walrasian equilibrium with price p^* . Suppose A^* is not Pareto optimal. Then $\exists A^{**} \in \bar{F}(\mathcal{E})$ such that

$$(i) \ u^i(x_{**}^i) \geq u^i(x_*^i), \forall i;$$

$$(ii) \ u^j(x_{**}^j) > u^j(x_*^j), \exists j.$$

Since u^i is strictly increasing, $\exists \epsilon > 0$ small enough that

$$u^j(x_{**}^j - (\epsilon, \dots, \epsilon)) > u^j(x_*^j).$$

Define

$$x_\epsilon^j := x_{**}^j - (\epsilon, \dots, \epsilon), \quad x_\epsilon^i := x_{**}^i + \frac{1}{I-1}(\epsilon, \dots, \epsilon), \quad i \neq j.$$

Then $u^i(x_\epsilon^i) > u^i(x_*^i)$ for all i , and

$$A^\epsilon = (x_\epsilon^1, \dots, x_\epsilon^I)$$

is feasible:

$$\sum_{i=1}^I x_\epsilon^i = \sum_{i=1}^I x_{**}^i = \sum_{i=1}^I \omega^i.$$

Since x_*^i solves consumer i 's problem and $u^i(x_\epsilon^i) > u^i(x_*^i)$,

$$p^* \cdot x_\epsilon^i > p^* \cdot \omega^i, \quad \forall i.$$

Summing over i ,

$$p^* \cdot \left(\sum_{i=1}^I x_\epsilon^i - \sum_{i=1}^I \omega^i \right) > 0,$$

a contradiction. Therefore A^* is Pareto optimal.

Definition 3.16. Production Economy

$$\mathcal{E} = \{(u^i, \omega^i)_{i=1, \dots, I}; Y_1, \dots, Y_J\}.$$

Aggregate supply:

$$\sum_{i=1}^I \omega^i + \sum_{j=1}^J y_j, \quad y_j \in Y_j.$$

$A = (x^1, \dots, x^I)$ is feasible if $\exists y_j \in Y_j$ such that

$$\sum_{i=1}^I x^i \leq \sum_{i=1}^I \omega^i + \sum_{j=1}^J y_j,$$

or equivalently,

$$\sum_{i=1}^I (x^i - \omega^i) \in \sum_{j=1}^J Y_j, \quad A + B = \{a + b \mid a \in A, b \in B\}.$$

$$F = \left\{ (x^1, \dots, x^I) \mid \sum_{i=1}^I (x^i - \omega^i) \in \sum_{j=1}^J Y_j \right\}.$$

Pure exchange: $\sum_{i=1}^I (x^i - \omega^i) \leq 0$.

$$U(\mathcal{E}) = \{(u^1(x^1), \dots, u^I(x^I)) \mid (x^1, \dots, x^I) \in F\}.$$

$A^* \in F$ is Pareto optimal iff

$$\{A \in F \mid u(A) > u(A^*)\} = \emptyset, \quad u(A) = \begin{bmatrix} u^1(x^1) \\ \vdots \\ u^I(x^I) \end{bmatrix}.$$

Proposition 3.20. Pareto Optimality in a Production Economy

$Y_j = \{y \in \mathbb{R}^L \mid F_j(y) \leq 0\}$. $A = (x^1, \dots, x^I) \in F$ is Pareto optimal iff $\exists y_j \in Y_j$ with $F_j(y_j) = 0$ such that

(i)

$$\frac{\partial u^i / \partial x_l^i}{\partial u^i / \partial x_k^i} = \frac{\partial F_j / \partial y_l}{\partial F_j / \partial y_k}, \quad \forall i, j, l, k;$$

(ii)

$$\sum_{i=1}^I x^i = \sum_{i=1}^I \omega^i + \sum_{j=1}^J y_j.$$

Definition 3.21. Private Ownership Economy

$$\mathcal{E} = \{(u^i, \omega^i, \theta^i)_{i=1, \dots, I}; Y_1, \dots, Y_J\}, \quad \theta^i = \begin{bmatrix} \theta_1^i \\ \vdots \\ \theta_J^i \end{bmatrix}, \quad \sum_{i=1}^I \theta_j^i = 1.$$

θ_j^i : household i 's share of firm j .

Definition 3.22. Competitive Equilibrium with Production

$A^* = (x_*^1, \dots, x_*^I; y_*^1, \dots, y_*^J)$ and $p^* \in \mathbb{R}_{++}^L$ constitute a competitive equilibrium if

(i)

$$y_*^j \in \arg \max_{y_j \in Y_j} p^* \cdot y_j;$$

(ii)

$$x_*^i \in \arg \max_{x^i} u^i(x^i) \quad \text{s.t.} \quad p^* \cdot x^i \leq p^* \cdot \omega^i + \sum_{j=1}^J \theta_j^i \pi_j^*, \quad \pi_j^* = p^* \cdot y_*^j;$$

(iii)

$$\sum_{i=1}^I x_*^i = \sum_{i=1}^I \omega^i + \sum_{j=1}^J y_*^j.$$

Example. Production Economy with Two Firms

$L = 4$.

$$u^i(x^i) = a \log x_1^i + (1 - a) \log x_2^i, \quad \omega^1 = (0, 0, 1, 0)^\top, \quad \omega^2 = (0, 0, 0, 1)^\top,$$

$$\theta^1 = \theta^2 = (1/2, 1/2)^\top, \quad y_j = 2\sqrt{z_3 z_4}, \quad j = 1, 2.$$

Firm j : cost minimization gives

$$z_3 = \frac{y_j}{2} \sqrt{\frac{p_4}{p_3}}, \quad z_4 = \frac{y_j}{2} \sqrt{\frac{p_3}{p_4}}, \quad C = y_j \sqrt{p_3 p_4}.$$

$$\pi_j = (p_j - \sqrt{p_3 p_4}) y_j.$$

CRS $\Rightarrow \pi_j^* = 0$:

$$p_1^* = p_2^* = \sqrt{p_3 p_4}.$$

Consumer demands ($\pi_1^* = \pi_2^* = 0$):

$$x_1^1 = a \frac{p_3}{p_1}, \quad x_2^1 = (1 - a) \frac{p_3}{p_2}, \quad x_1^2 = a \frac{p_4}{p_1}, \quad x_2^2 = (1 - a) \frac{p_4}{p_2}.$$

Market clearing:

$$a \frac{p_3 + p_4}{p_1} = y_1, \quad (1 - a) \frac{p_3 + p_4}{p_2} = y_2,$$

$$\frac{y_1 + y_2}{2} \sqrt{\frac{p_4}{p_3}} = 1, \quad \frac{y_1 + y_2}{2} \sqrt{\frac{p_3}{p_4}} = 1.$$

With $p_1 = p_2 = 1$:

$$\begin{cases} a(p_3 + p_4) = y_1, \\ (1 - a)(p_3 + p_4) = y_2, \\ \frac{y_1 + y_2}{2} \sqrt{\frac{p_4}{p_3}} = 1, \\ \frac{y_1 + y_2}{2} \sqrt{\frac{p_3}{p_4}} = 1. \end{cases}$$

$$\Rightarrow p_3 = p_4 = p, \quad y_1 = 2ap, \quad y_2 = 2(1 - a)p, \quad p_1^* = p_2^* = p.$$

Setting $p = 1$:

$$p_1^* : p_2^* : p_3^* : p_4^* = 1 : 1 : 1 : 1,$$

$$y_1 = 2a, \ z_3 = a, \ z_4 = a, \quad y_2 = 2(1 - a), \ z_3 = 1 - a, \ z_4 = 1 - a,$$

$$x_1^1 = x_1^2 = a, \quad x_2^1 = x_2^2 = 1 - a.$$

$$A^* = \left(\left(\begin{pmatrix} a \\ 1 - a \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} a \\ 1 - a \\ 0 \\ -1 \end{pmatrix} \right); \begin{pmatrix} 2a \\ 0 \\ -a \\ -a \end{pmatrix}, \begin{pmatrix} 0 \\ 2(1 - a) \\ -(1 - a) \\ -(1 - a) \end{pmatrix} \right).$$